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1 May 1979
SUPERSEDING
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25 August 1971

MILITARY SPECIFICATION

WATER PURIFICATION UNIT, FRAME MOUNTED DIATOMITE TYPE, 25 GPM

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gasoline-engine-driven, two-section, frame mounted diatomaceous earth-type, water purification unit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-387 - Plastic Sheet, Laminated, Thermosetting (for Designation Plates).
FF-B-575 - Bolts, Hexagon and Square.
FF-N-836 - Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
TT-P-664 - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4610

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WW-C-624 - Coupling Assembly, Hose (Garden, Water, and Water Suction).
CCC-D-950 - Dyeing and Aftertreating Processes for Cotton Cloths.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical, Packaging of.
MIL-C-538 - Comparator, Color.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-T-22085 - Tape, Pressure-Sensitive Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED STD H28 - Screw-Threads Standards for Federal Services.
FED STD 595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

NAVAL FACILITIES ENGINEERING COMMAND

605203 - Machinery Section.
605204 - Reservoir and Tool Box Details.
605205 - Restrictor and Sight Glass Details.
605206 - Filter Section Assembly and Details.
605207 - Filter Details.
605208 - Flow Controller.
605209 - Instruction and Identification Plate Details.
605210 - Slurry Feeder Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The purification unit shall include a gasoline engine, pumping equipment, filter aid (diatomaceous earth) feeder, hypochlorite feeder, and filtering apparatus. All equipment, piping, and accessories specified herein shall be included.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete water purification unit for first article inspection and approval (see 4.3 and 6.4).

* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial

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practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by the specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under the specification unless otherwise specifically specified.

* 3.4 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.5 Design and construction. The water purification unit shall be so constructed as to be divisible into two sections, a machinery section and a filter section, each transportable by motor vehicle. The uncrated weight of the machinery section shall not exceed 450 pounds (lb). The uncrated weight of the filter section shall not exceed 300 lb, excluding accessories. When discrepancies exist between the specifications and drawings, the specification shall govern. Minor changes in design from those shown on the drawings may be subject to approval of the contracting officer at time of submission of the first article. All screw threads shall conform to FED STD H28.

3.6 Machinery section. The machinery section shall be in accordance with Drawing 605203. The section shall consist of the pump, gasoline engine, hypochlorite feeder, hypochlorite reservoir, filter and slurry feeder with hopper, tool box, necessary piping and valves, canvas coverall, and other accessory equipment specified herein.

* 3.6.1 Pump and engine. The pump shall be a single-stage, centrifugal, integral self-priming-type, direct-connected to a gasoline engine. The engine and pump shall be mounted on a common steel base. When operated at not more than 2,200 revolutions per minute (rpm), the unit shall be capable of delivering continuously a minimum of 55 gallons per minute (gpm) against a total dynamic head of 50 feet, including 15 feet of suction lift. The pump shall prime in not more than 4 minutes when drafting water through a 1-1/2 inch suction hose with a suction lift of 15 feet. The pump housing, including the packing, shall not leak while the pump develops maximum shut-off pressure. The engine shall be a single-cylinder, 4-cycle, air-cooled gasoline engine, capable of operating the pump continuously at the head and capacity specified

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without exceeding the net continuous brake horsepower rating of the engine. The engine shall be provided with a rope starter and an auxiliary power takeoff through a V-belt pulley mounted on the crank shaft and a fuel tank having a capacity of not less than 1 gallon. The pump and engine base shall be provided with four shock mounts for effective isolation of vibration.

* 3.6.1.1 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions indicated in this specification.

3.6.1.2 Engine shock mount. The shock mount shall consist of an oil-resistant rubber compound having a low compression set and a 75 ±5 Shore A durometer hardness. A hole in the center of the mount shall provide access for a 3/8-inch bolt.

3.6.1.3 Bolts and nuts. Bolts shall conform to type 2 of FF-B-575. Nuts shall conform to type II, style I of FF-N-836.

3.6.1.4 Attachment. The engine and the pump shall be securely attached to the common steel base by the shock mounts and by nuts and bolts. The attachment must be capable of eliminating vibrations.

3.6.2 Hypochlorite feeder. The hypochlorite feeder shall be a diaphragm pump driven by the slurry feeder shaft. The feeder shall have a pumping range of from 14 to 56 gallons of a 1 percent hypochlorite solution per 24 hours when driven at a speed of 600 rpm. The feeder shall be capable of operating successfully against pressures of from 0 to 30 pounds per square inch gage (psig). The pumping head assembly shall consist of a hard rubber or plastic body; a fabric reinforced diaphragm; suction and discharge valves of the poppet type; and a manually operated vent. The assembly shall be mounted on a cast iron or aluminum housing containing the gear reduction mechanism and the diaphragm reciprocating mechanism, both of which shall operate in an oil bath. A pumping stroke adjuster, with indicating dial, operable when the feeder is in operation shall be included.

3.6.3 Hypochlorite reservoir. The body of the hypochlorite reservoir shall be a one-piece molded container of 1/4-inch thick acrylonitrile, butadiene, and styrene plastic with reinforced edges similar to that shown on Drawing 605204. A cover made of 1/8-inch plastic sheet shall be furnished with the reservoir. Provisions shall be made for securing the cover in place and yet provide for easy removal

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3.14 Identification marking. The machinery section, engine, and filter section shall be marked for identification in accordance with Drawing 605209.

3.15 Drawings. Unless the equipment offered is identical to that shown on the drawings specified herein, the bidder shall submit with his bid, general arrangement and assembly drawings sufficiently detailed to permit clear understanding of the design of the unit and its component parts.

* 3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.16.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of a uniform shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.16.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses wwithout permanent deformation or failure when subjected to proof or service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings

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